

Extra Task

Name : Shivam Ramakant Deshmane

Domain : AI & ML

College : T.B Girwalkar Polytechnic Ambajogai

[2]

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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

[5]
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▶ `df = pd.read_csv("train.csv")`

[6]
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```
print(df.head())
```

▼

```
PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                Name    Sex  Age  SibSp  \
0                        Braund, Mr. Owen Harris    male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2                        Heikkinen, Miss. Laina  female  26.0    0
3      Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1
4                        Allen, Mr. William Henry    male  35.0    0

    Parch    Ticket    Fare Cabin Embarked
0      0    A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0    113803   53.1000  C123      S
4      0    373450   8.0500   NaN      S
```



```
(891, 12)
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
PassengerId    int64
Survived       int64
Pclass         int64
Name           object
Sex            object
Age            float64
SibSp          int64
Parch          int64
Ticket         object
Fare           float64
Cabin          object
Embarked       object
dtype: object
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
```

```

Ticket      0
Fare        0
... Cabin    687
Embarked     2
dtype: int64

   PassengerId  Survived  Pclass     Age  SibSp  \
count  891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std    257.353842    0.486592    0.836071   14.526497    1.102743
min      1.000000    0.000000    1.000000    0.420000    0.000000
25%    223.500000    0.000000    2.000000   20.125000    0.000000
50%    446.000000    0.000000    3.000000   28.000000    0.000000
75%    668.500000    1.000000    3.000000   38.000000    1.000000
max    891.000000    1.000000    3.000000   80.000000    8.000000

      Parch     Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std     0.806057   49.693429
min     0.000000    0.000000
25%     0.000000    7.910400
50%     0.000000   14.454200
75%     0.000000   31.000000
max     6.000000  512.329200

```

[9]

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```
print(df.tail())
```

▼

	PassengerId	Survived	Pclass	Name \
886	887	0	2	Montvila, Rev. Juozas
887	888	1	1	Graham, Miss. Margaret Edith
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"
889	890	1	1	Behr, Mr. Karl Howell
890	891	0	3	Dooley, Mr. Patrick

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	male	27.0	0	0	211536	13.00	NaN	S
887	female	19.0	0	0	112053	30.00	B42	S
888	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	male	26.0	0	0	111369	30.00	C148	C
890	male	32.0	0	0	370376	7.75	NaN	Q

[10]

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```
print(df.columns)
```

▼

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',  
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],  
      dtype='object')
```

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df.info()



```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   PassengerId     891 non-null   int64
 1   Survived        891 non-null   int64
 2   Pclass          891 non-null   int64
 3   Name            891 non-null   object
 4   Sex             891 non-null   object
 5   Age             714 non-null   float64
 6   SibSp           891 non-null   int64
 7   Parch           891 non-null   int64
 8   Ticket          891 non-null   object
 9   Fare            891 non-null   float64
10   Cabin           204 non-null   object
11   Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```


[12]

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```
print(df.dtypes)
```



```
... PassengerId    int64
      Survived      int64
      Pclass       int64
      Name         object
      Sex          object
      Age          float64
      SibSp         int64
      Parch        int64
      Ticket       object
      Fare         float64
      Cabin        object
      Embarked     object
dtype: object
```

[13]

✓ 0s



```
print(df.isnull().sum())
```



```
... PassengerId      0
      Survived       0
      Pclass        0
      Name          0
      Sex           0
      Age          177
      SibSp         0
      Parch         0
      Ticket        0
      Fare          0
      Cabin        687
      Embarked       2
      dtype: int64
```

[14]

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 `print(df.describe())`



...

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

[15]

✓ 0s



```
print(df.nunique())
```



```
... PassengerId    891
   Survived        2
   Pclass          3
   Name           891
   Sex             2
   Age            88
   SibSp           7
   Parch           7
   Ticket         681
   Fare           248
   Cabin          147
   Embarked        3
dtype: int64
```

✓ 0s

person/

▼

...

Fare Distribution

